DEPARTMENT OF ELECTRICAL ENGINEERING

Sub: ADE (EEPC202), B.Tech 3rd sem, Time: 50 min, MM: 15

Q.1	For the emitter-bias network , $V_{CC} = 20\,V, R_B = 430k\Omega, R_C = 2k\Omega, R_E = 1k\Omega, \beta = 50$ Determine $(a)\ I_B\ \ (b)\ I_C\ \ (c)\ V_{CE}\ (d) V_C\ \ (e)\ V_E\ \ (f)\ V_B\ \ (g)\ V_{BC}$	5
Q.2	Derive the input impedance, output impedance, and voltage gain for the voltage divider bias common emitter BJT configuration.	5
Q.3	Convert the following decimal numbers to the indicated bases: (a) $(7562.45)_{10}$ to $()_8$ (b) $(1938.257)_{10}$ to $()_{16}$ → 792 (c) $(175.175)_{10}$ to $()_2$ → 10101111 (d) $(124)_5$ to $()_2$ (e) $(712)_8$ to $()_{10}$ → 458	5